



JAIPUR NATIONAL UNIVERSITY

End Semester/Year Examination December 2025

Name of School:	School of Engineering & Technology
Name of Programme / Course:	B.Tech (Computer Science Engineering / AIML / CY / Data Science)
Semester/Year:	V / III
Paper Code:	BTECSCO504T22
Title of Paper:	Programming in Java
Time: 3 Hrs.	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

1. This question paper consists of two sections, Section-A and Section-B.
 2. Section-A consists of seven questions and
 3. Section-B consists of six questions.
- Attempt any five questions from Section-A and any four questions from Section-B. Marks allotted are mentioned against each question

Question Number	SECTION - A	Marks	Bloom Level	Course Outcome
Q.1	Differentiate between Java and C++ in terms of memory management, platform independence, and security.	<u>5</u>	2	1
Q.2	Explain the concept of Bytecode and JVM. Why is Java called a platform-independent language?	5	2	1
Q.3	What is the purpose of the this and super keywords? Give an example to illustrate their use.	5	3	2
Q.4	Write short notes on Interfaces and Abstract Classes with suitable examples.	5	2	3

Q.5	What are checked and unchecked exceptions? Give examples and explain the use of try, catch, and finally.	5	3	3
Q.6	Write a short note on Serialization and explain its purpose in Java file handling.	5	1	4
Q.7	Differentiate between AWT and Swing components. Give one example where Swing provides an advantage.	5	4	5

SECTION - B

10

Q.1 (a)	Explain the features and advantages of Java over traditional programming languages.	10	2 & 4	1
(b)	Discuss the use of Java class libraries <u>java.lang</u> and <u>java.util</u> with examples.			
Q.2 (a)	Explain Inheritance in Java with an example showing method overriding .	10	3	2
(b)	Write a program to demonstrate multithreading by creating multiple threads performing different tasks concurrently.			

Q.3 (a)	Explain package creation and class importing with suitable syntax and example.	10	3 & 4	3
(b)	Write a program to define and use a custom exception class for invalid user input.			

Q.4 (a)
(b)

Q.5 (a)
(b)

Q.6 (a)
(b)

3 3

1 4

4 5

2 & 4 1

3 2

3 & 4 3

~~Q.4~~ (a)

Explain the process of **reading and writing files** using the java.io package.

10

3

4

(b)

Write a Java program to **copy the contents of one file to another** using file streams.

10

3 & 4

4

~~Q.5~~ (a)

Write a Java applet to draw **basic geometric shapes (circle, rectangle, and line)** using the Graphics class.

(b)

Discuss how to handle **image and audio files** in Java multimedia applications.

10

6 & 2

5

~~Q.6~~ (a)

Design a **Swing-based GUI application** that takes user input and displays it dynamically in a text area using **event handling**.

(b)

Explain the **Delegation Event Model** and its components in Java.